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First-Order Logic:

First-Order Logic (FOL), or predicate logic, is a powerful knowledge representation tool in AI. It expands simple true/false statements into structured models of objects, properties, and relationships, allowing machines to perform complex logical reasoning. Commonly applied in NLP, expert systems, and theorem proving.

Use of FOL in AI

FOL is useful for **knowledge representation and reasoning**. AI systems can use it to represent facts about a domain and infer conclusions from those facts. It has applications in expert systems, automated theorem proving, semantic reasoning, planning, and some forms of natural-language understanding.

In short: First-Order Logic provides AI with a precise way to represent **objects, properties, relationships, and rules**, enabling the system to reason logically from known information.

Core Components

- **Constants:** Specific names or things (like John or NewYork).
- **Variables:** Placeholders for any object (like x or y).
- **Predicates:** Actions, traits, or relations (like Brother(John, Bob)).
- **Functions:** Mappings from one object to another (like FatherOf(John)).
- **Quantifiers:** Rules defining object scope, such as Universal (∀, meaning "for all") and Existential (∃, meaning "there exists").
- **Logical Connectives**: Include conjunction(∧∧), disjunction (∨), implication (→), biconditional (↔) and negation (¬).

Example of Reasoning

- Fact 1: $\forall x$(Human(x) →Mortal(x)) — "All humans are mortal."
- Fact 2: Human (Socrates) — "Socrates is a human."
- Inference: Mortal (Socrates) — "Socrates is mortal."

Applications

- **Knowledge Representation:** Represents objects, properties, and relationships in a structured form for reasoning tasks
- **Automated Theorem Proving:** Applies logical rules to prove mathematical statements and verify system correctness
- **Natural Language Processing (NLP):** Helps convert natural language into logical representations for understanding and reasoning
- **Expert Systems:** Encodes domain knowledge to support decision-making in systems like medical or legal assistants
- **Semantic Web:** Defines relationships between web resources for improved search and information retrieval.

Problem 1:

Suppose we know:

Fact: Socrates is a human.

$$Human(Socrates)$$

Rule: All humans are mortal.

$$\forall x \ (Human(x) \rightarrow Mortal(x))$$

From these, the AI can infer:

$$Mortal(Socrates)$$

Problem 2: Students and courses

Suppose:

- Rahim is a student.
- Every student studies at least one course.
- Therefore, Rahim studies a course.

FOL representation:

Fact: Student (Rahim)

Rule: ∀x [Student(x)→∃y Studies(x,y)]

Therefore: Student(Rahim)→∃y Studies(Rahim,y)
Since: Student(Rahim)=True

We conclude: ∃y Studies(Rahim, y)

Meaning: **There exists at least one course that Rahim studies.**

Problem 3: Family relationship

Suppose: Everyone's father is a parent.

FOL:

∀x∀y [Father(y,x)→Parent(y,x)]

Given:

Father(Ali,Rahim)

Then:

Father(Ali,Rahim)→Parent(Ali,Rahim)

Therefore:

Parent(Ali,Rahim)

Problem 4: Using AND (∧)
Suppose an AI system knows:
If a person is a student **and** studies AI, then the person is an AI student.

FOL:
$\forall x[(Student(x) \wedge Studies(x,AI)) \rightarrow AIStudent(x)]$

Given:
Student(Tanvir)
and
Studies(Tanvir, AI)
Therefore:
Student(Tanvir)∧Studies(Tanvir,AI)
is **True**, so:
AIStudent(Tanvir)

Problem 5: Using NOT (¬)
Suppose:
No student is a professor.
FOL: $\forall x[Student(x) \rightarrow \neg Professor(x)]$
Given: Student (Rahim)
Therefore: ¬Professor(Rahim)
So the AI can conclude that **Rahim is not a professor**.

Problem 6: A typical AI reasoning problem
Knowledge Base:
Human(Ali)
Human(Rahim)
$\forall x[Human(x) \rightarrow Mortal(x)]$
Question: Is Ali mortal?
Substitute x=Al:
$Human(Ali) \rightarrow Mortal(Ali)$
Since: Human(Ali)=True
We infer: Mortal(Ali)
Similarly, because:
Human(Rahim)=True
We infer: Mortal(Rahim)
Important FOL pattern for exams
A very common structure is: $\forall x(P(x) \rightarrow Q(x))$
Given: P(a) Infer: Q(a)
For example: $\forall x(Student(x) \rightarrow Intelligent(x))$
Student(Ali)
Therefore: Intelligent(Ali)
This is essentially **Universal Instantiation + Modus Ponens**, and it is one of the most important reasoning patterns in FOL for AI.